

PENETRATION TESTING REPORT

FOOTPRINTING, OSINT & NETWORK SCANNING PHASES

W2-PM1, W2-PM3, W2-PM4, W2-PM5 | CYBERSECURITY | NETWORKWALKS

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Program/Batch	B082-Networkwalks
Date	18 September 2026
Modules completed	W2-PM1 (Multiple Kali Tools) W2-PM3 (Maltego) W2-PM4 (The Harvester) W2-PM5 (Zenmap Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using multiple Kali Linux tools (W2-PM1), OSINT enumeration and visualization using theHarvester and Maltego (W2-PM3, W2-PM4), and scanning my own local network with Zenmap (W2-PM5). Together these modules show how an attacker moves from gathering public information to mapping live hosts on a network. It is part of my ongoing internship program at Networkwalks.

All footprinting and OSINT commands were run in Kali Linux, and the network scan was run on a Windows PC with Zenmap installed. Every step below includes the exact tool used, the result observed, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
Kali Linux & Windows	Operating systems used for reconnaissance and scanning activities
whois	Find domain registration details: owner, dates, name servers
whatweb	Fingerprint web technologies: server, CMS, plugins, IP
nslookup	Resolve the domain name to its IP address using DNS
curl -I	Read the HTTP response headers of the website
wafw00f	Detect whether a Web Application Firewall protects the site
dnsrecon	Enumerate DNS records: NS, MX, SPF, TXT, SRV
theHarvester	Enumerate subdomains, hosts, IPs and emails from public sources
Maltego	Visually map relationships between a domain, its infrastructure and associated entities in a single OSINT graph
Zenmap (Nmap GUI)	Scan the local subnet to find live hosts, open ports, services and operating systems
Windows CMD	Local IP, subnet and MAC address identification using ipconfig

4. Activities Performed

4.1 Footprinting & Reconnaissance

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: **WHOIS**, **WhatWeb**, **Nslookup**, **Curl**, **Wafw00f** and **DNSRecon**. Each tool was used to collect a different type of information about the target.

First, I used **WHOIS** to obtain publicly available domain registration information and identify the domain's name servers. The results identified the registrar as GoDaddy, a creation date of 2019-11-06, an expiry date of 2027-11-06, and hosting on HostGator (NS6135/6136.hostgator.com), with registrant privacy provided through DomainsByProxy.

I then used **WhatWeb** to identify technologies used by the website. The results identified **WordPress 7.1** and **WP Download Manager 3.3.58**, along with jQuery 3.7.1 and Bootstrap 7.1.

Using **Nslookup**, I resolved the domain name to its IP address. The result identified **192.232.216.135**.

I used **Curl** with the -I option to inspect the HTTP response headers. This showed a referrer-policy of no-referrer-when-downgrade and a permissions-policy header, with no CSP or HSTS header observed.

Next, I used **Wafw00f** to determine whether a Web Application Firewall was protecting the website. The result identified **ModSecurity (SpiderLabs)**.

Finally, I used **DNSRecon** to enumerate DNS records. The results provided information relating to name servers, an MX record pointing to mail.networkwalks.com, an SPF/TXT record, a google-site-verification record, and autodiscover SRV records used for cPanel email client configuration.

4.2 OSINT Enumeration & Visualization

For this activity I used **theHarvester** and **Maltego** to demonstrate large scale OSINT enumeration and to consolidate the collected data visually.

I ran **theHarvester** against microsoft.com (using the -b all option) to demonstrate its enumeration capability on a large public organization. The results returned approximately 9,966 hosts, 48 IPs and 5 generic, role based email addresses such as postmaster@ and secure@, reflecting the size of microsoft.com's public footprint rather than a specific weakness.

I then used **Maltego** to visually map the domain. I added networkwalks.com as a Domain entity and ran the "To Email Addresses [Search Engine]" transform, which identified the email address **info@networkwalks.com** linked to the domain. This showed how Maltego can take a single starting entity and consolidate related public data into a visual OSINT graph.

4.3 Network Scanning with Zenmap

For the third activity, I used **Zenmap** to perform network discovery on my local network. I entered the subnet **192.168.89.0/24** into Zenmap and ran an **Intense scan** profile, which executed the command **nmap -T4 -A -v 192.168.89.0/24**.

The scan checked 256 IP addresses and completed in 46.53 seconds, identifying two live hosts: **192.168.89.98** and **192.168.89.170**.

For host 192.168.89.98, the scan identified four open TCP ports: 135/tcp (msrpc, Microsoft Windows RPC), 139/tcp (netbios-ssn, Microsoft Windows netbios-ssn), 445/tcp (microsoft-ds), and 5357/tcp (wsdapi). The host was fingerprinted as **Microsoft Windows 10 (build 1809 to 21H2)**, with a network distance of 0 hops. The host script results showed SMB2 security mode 3.1.1 with message signing enabled but not required.

After completing the scan, I opened the Topology section in Zenmap and generated a network diagram showing both live hosts connected to the scanning machine (localhost).

I also used **Command Prompt** (ipconfig) on the Windows scanning machine to confirm local network configuration, including the local IPv4 address 192.168.89.170 and its physical (MAC) address FE-F0-5C-11-9C-B3. Combining this with the Nmap/Zenmap results, the MAC addresses for both live hosts were identified as follows: **192.168.89.98** → AA:E4:77:0F:07:5F, and **192.168.89.170** → FE:F0:5C:11:9C:B3 (the scanning machine itself).

5. Risk Analysis / Impact

Based on the information collected during the footprinting, OSINT and network scanning activities, I identified the following potential risks.

#	Risk / Finding	Evidence / Observation	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb identified WordPress and WP Download Manager with version numbers	Attackers may use exposed technology/version information to identify software requiring further security review	Medium
2	Server IP address identifiable	Nslookup resolved the domain to 192.232.216.135	Provides information about the network location of the web service	Low
3	DNS infrastructure information exposed	DNSRecon identified DNS, mail and service related records	DNS information can help build a broader infrastructure profile	Medium
4	Large public OSINT footprint	theHarvester identified thousands of public hosts and emails for a large organization	A large attack surface increases the number of possible entry points for an attacker to investigate	Medium
5	SMB signing not enforced	Zenmap/Nmap smb2-security-mode script showed message signing enabled but not required on 192.168.89.98	A host without enforced SMB signing may be vulnerable to relay or man in the middle style attacks on the local network	Medium
6	Multiple live hosts visible on local network	Zenmap and Command Prompt identified two live hosts with MAC addresses AA:E4:77:0F:07:5F (192.168.89.98) and FE:F0:5C:11:9C:B3 (192.168.89.170) on the scanned subnet	Unknown or unauthorized devices may potentially be present on a network	Low

The risks above are observations from the footprinting, OSINT and scanning exercises, not confirmed vulnerabilities. These activities primarily involved information gathering and host discovery; no exploitation or vulnerability validation was performed. Therefore, the presence of a software version, IP address, DNS record or open port does not by itself mean that a system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

- 1. Review publicly exposed technology information:** Regularly review what information about web technologies, CMS and plugins is publicly visible.
- 2. Keep software updated:** CMS platforms, plugins and other web technologies should be regularly updated and reviewed against current security advisories.
- 3. Review HTTP headers:** HTTP response headers should be reviewed to determine whether unnecessary technical information is being exposed.
- 4. Review DNS records regularly:** DNS records should be checked periodically to ensure only required information and services are publicly exposed.

5. **Properly configure and monitor the WAF:** Keep the WAF (ModSecurity) enabled and tuned, since it already blocks naive attacks.
6. **Limit public OSINT exposure:** Periodically audit publicly discoverable subdomains and email addresses to reduce the organization's exposed footprint.
7. **Enforce SMB signing:** Configure Windows hosts to require SMB signing rather than only enabling it, to reduce the risk of relay attacks.
8. **Perform regular internal network discovery:** Periodically scan internal networks to identify active devices.
9. **Investigate unknown devices:** Any unexpected device discovered during network scanning should be investigated and verified.
10. **Maintain network documentation:** Network topology and device information should be documented and updated regularly.
11. **Perform security testing with authorization:** Reconnaissance and scanning should only be performed against systems and networks where appropriate authorization has been provided.

7. Conclusion

During this phase of my Cybersecurity & Ethical Hacking internship, I completed practical activities covering footprinting, OSINT enumeration and network scanning.

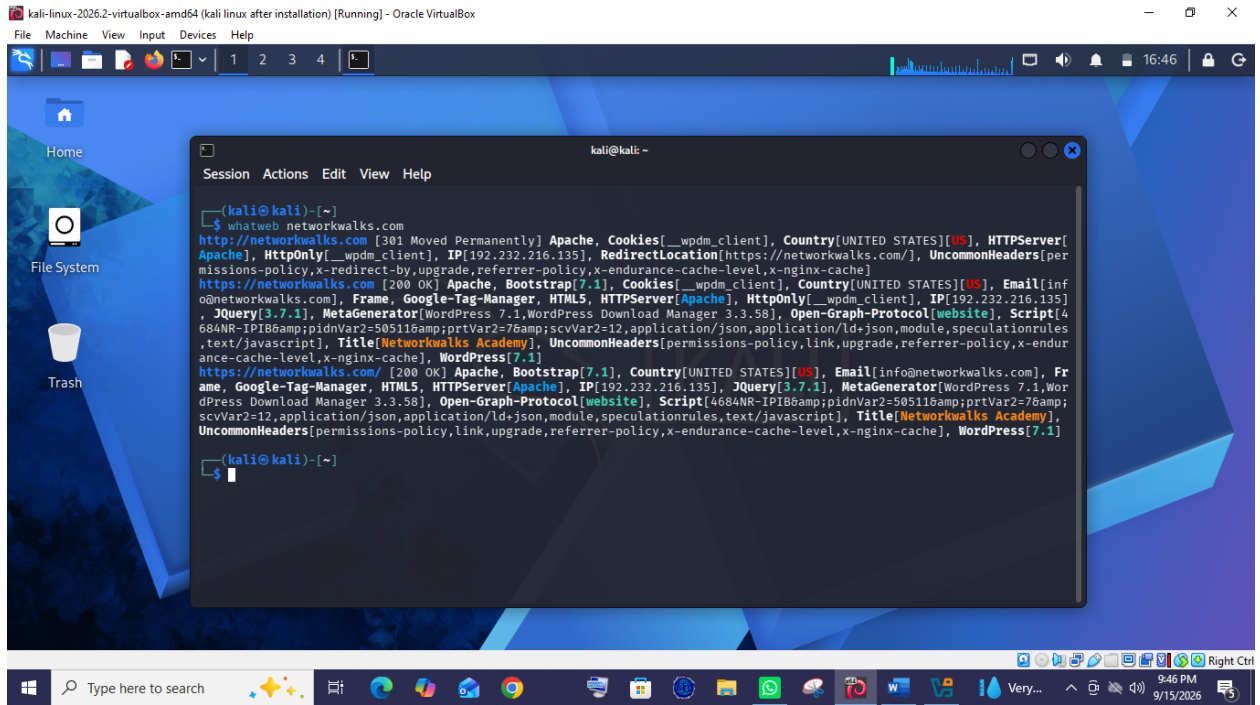
In the footprinting activity, I used six Kali Linux tools to collect information about the target domain. I learned how WHOIS can provide domain information, WhatWeb can identify web technologies, Nslookup can resolve domain names, Curl can inspect HTTP headers, Wafw00f can identify a WAF, and DNSRecon can provide additional DNS information.

In the OSINT activity, I used theHarvester to demonstrate large scale public data enumeration and Maltego to visually consolidate the collected information into a single graph, showing how scattered technical data can be turned into a readable picture of an organization's footprint.

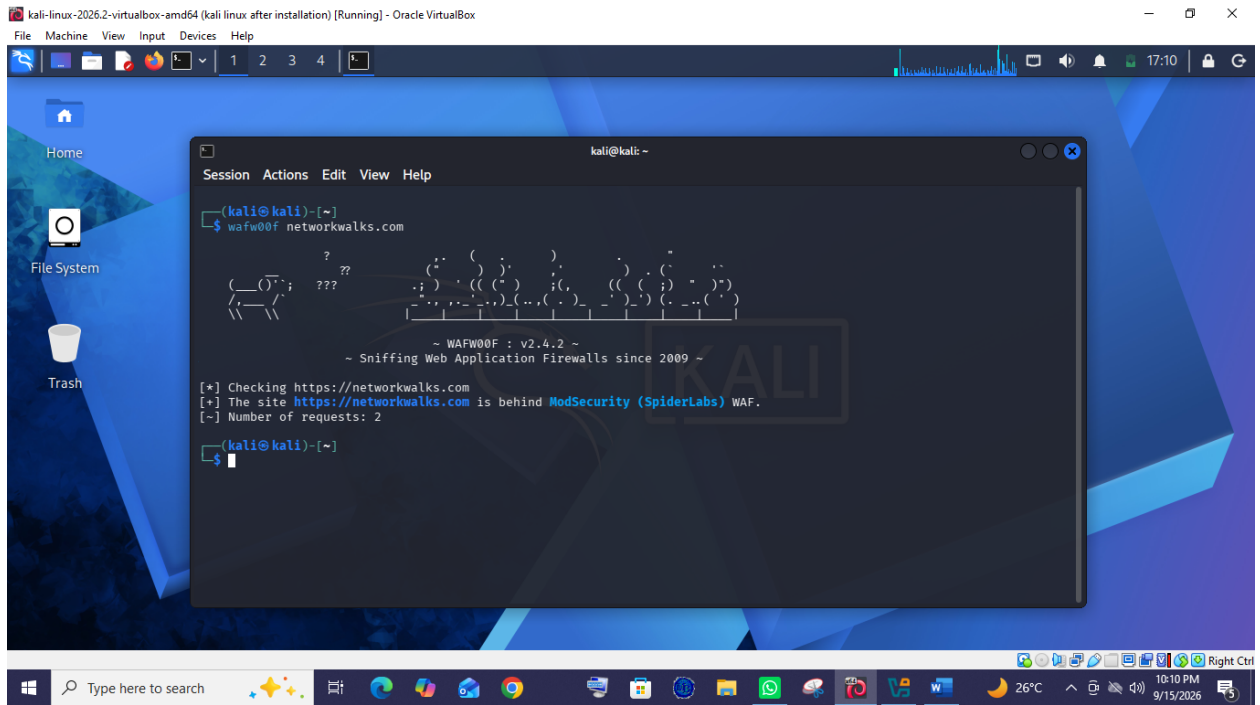
In the network scanning activity, I used Zenmap to scan my local subnet, identify live hosts, open ports and an operating system fingerprint, and to generate a network topology diagram.

The exercises showed me that information gathering is an important part of cybersecurity. Even before attempting to exploit a system, a security professional can learn a significant amount about an environment by carefully analyzing publicly available information and network responses. I also learned that reconnaissance and scanning must always be performed within an authorized scope, as these activities were completed as part of the assigned educational cybersecurity lab.

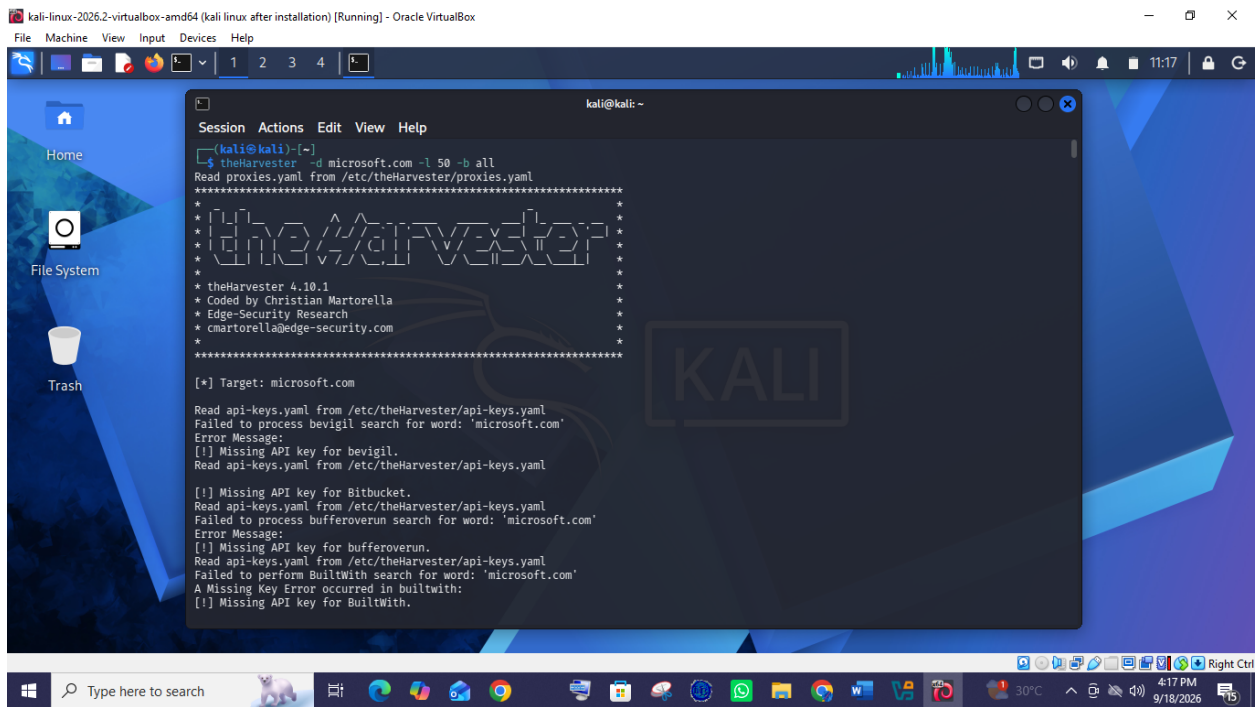
8. Evidences Collected



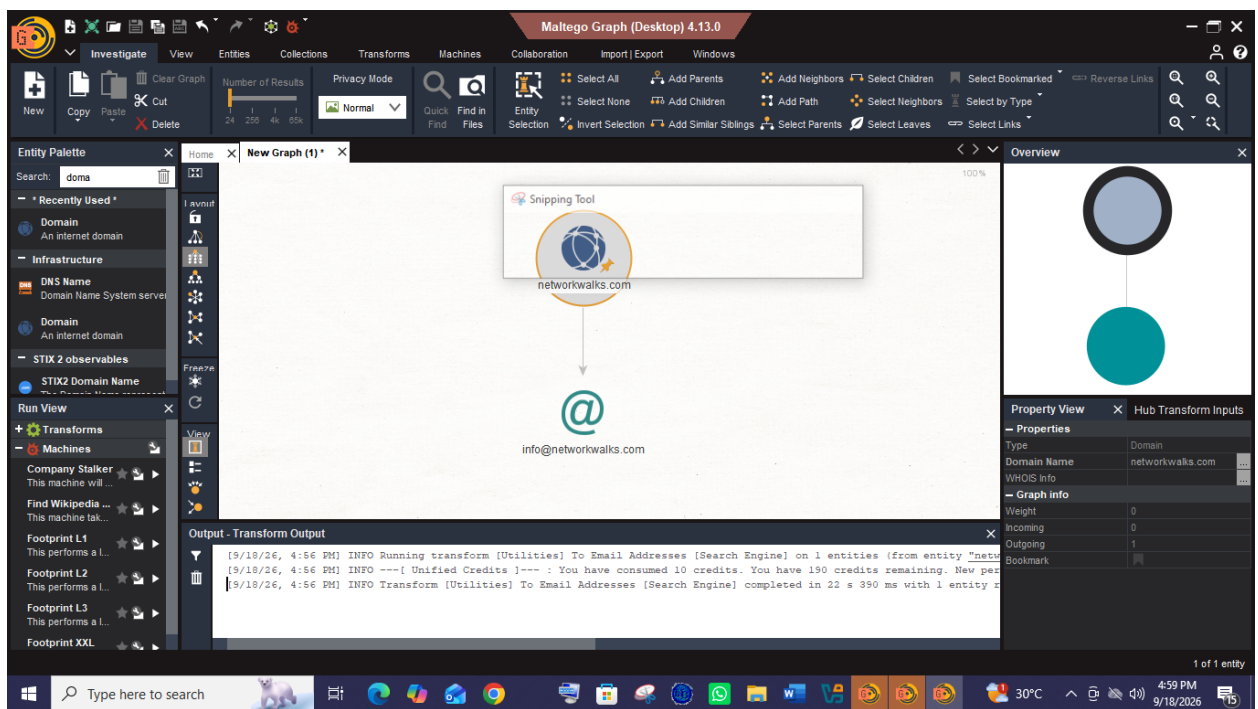
WhatWeb output for networkwalks.com identifying WordPress, Bootstrap and jQuery versions



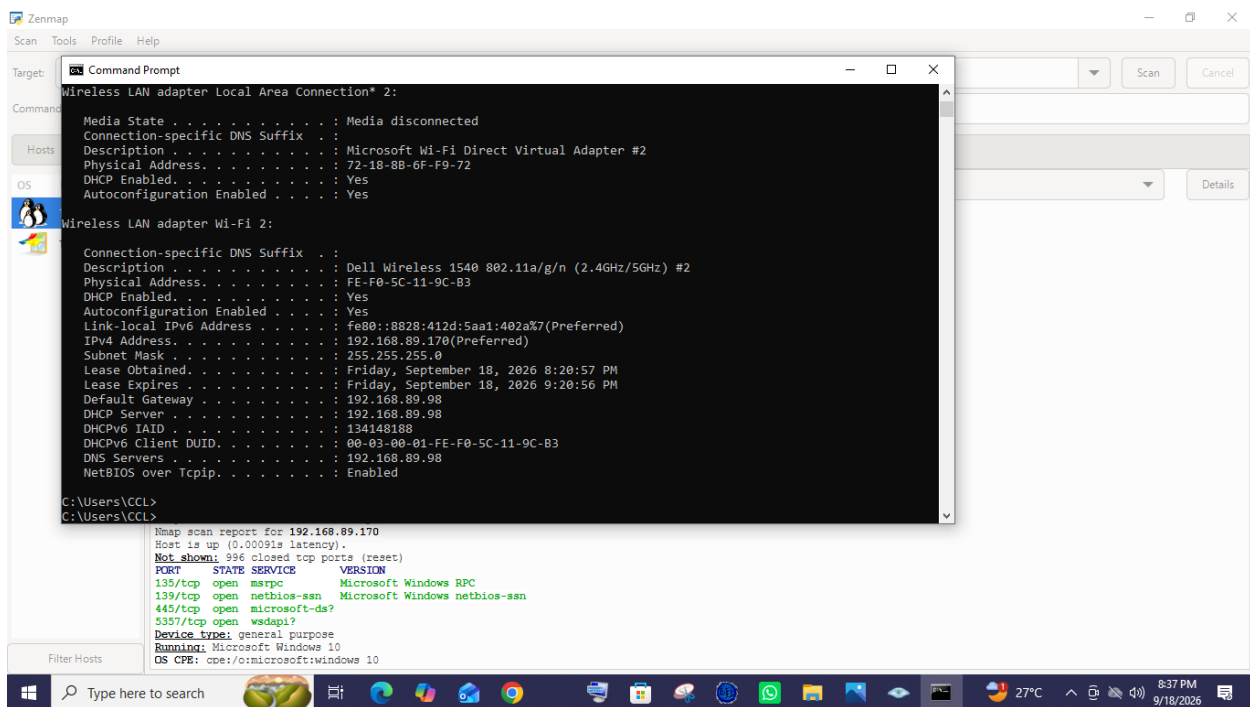
Wafw00f output identifying ModSecurity (SpiderLabs) protecting networkwalks.com



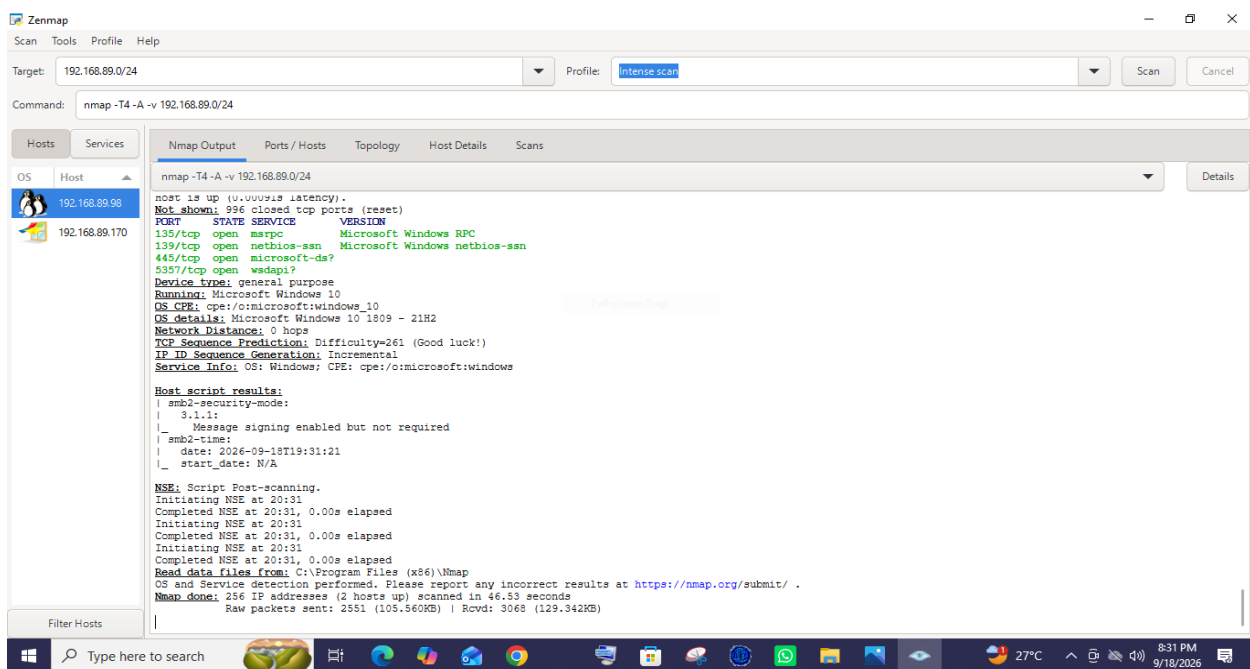
theHarvester execution against microsoft.com



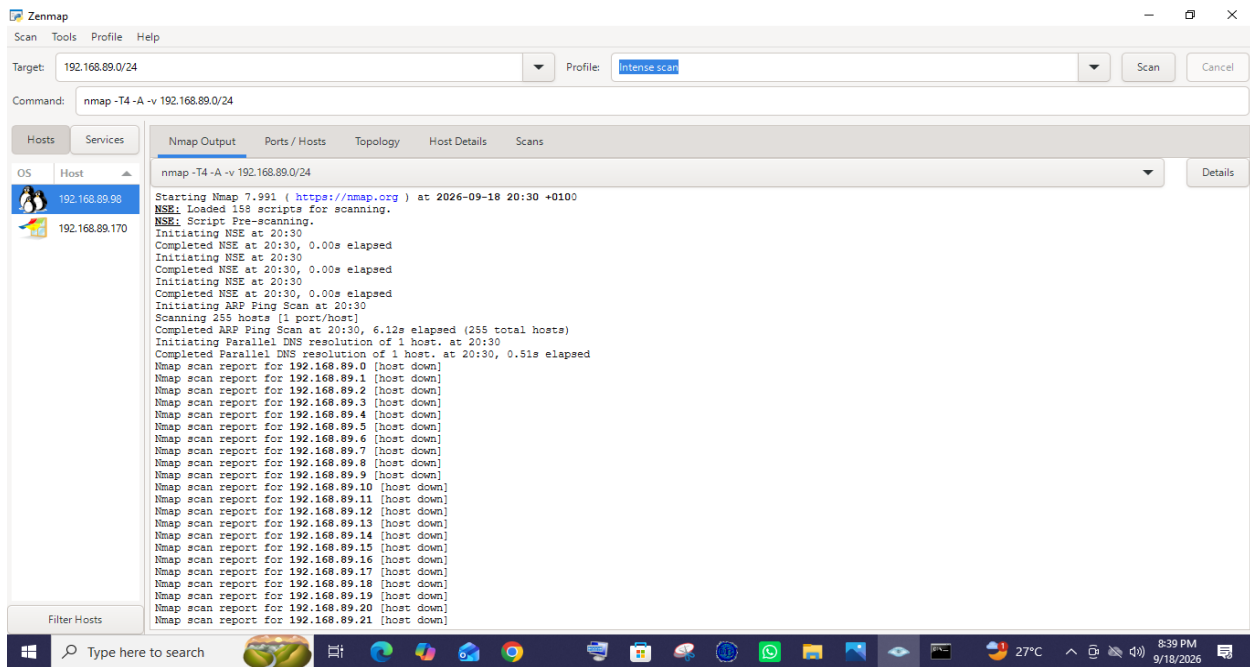
Maltego graph showing networkwalks.com domain entity linked to info@networkwalks.com via the To Email Addresses transform



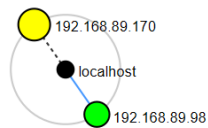
Command Prompt (ipconfig) output showing local IP and MAC address of the scanning machine



Zenmap Intense scan output, host 192.168.89.98 detail



Zenmap Intense scan output, scan initiation and host discovery



Zenmap network topology showing live hosts